

Programming Fundamentals - Topic 03 Tutorial

Tutorial Questions - Variables, Data Types & Operators

Concept Check

1. What is a variable in Java, and why do we need to specify its data type?
2. Give two examples of valid variable names and two invalid ones. Explain why.

Keywords

3. What is a *keyword* (reserved word) in Java?
4. Why can't you name a variable `class`?

Tokens

5. Break the line below into its tokens and identify their roles:

```
double price = 19.99;
```

Primitive Data Types

6. What is the difference between `int` and `double`?
7. When would you choose a `boolean` instead of an `int`?

Errors

8. Why is `Int x = 5;` invalid?
9. What happens if you try `int x = 5.5;`?

Operators

10. What is the difference between `/` when used with integers vs. doubles?
11. Predict the output of:

```
int a = 10, b = 6;  
float c = 6f;  
System.out.println(a / b);  
System.out.println(a / c);
```

Discussion

12. Why might `gearRatio` be a better variable name than `gr`?
13. How does using clear variable names make it easier to read a program and fix any logical errors?

Tutorial Questions -- Local Variables & Scope

Local Variables

1. How are local variables different from fields?
2. When are local variables created and destroyed?

Scope

3. What do we mean by the *scope* of a variable?
4. Explain why the following will not compile:

```
{  
    int x = 5;  
}  
System.out.println(x);
```

Refund Example

5. If we wish the `refundBalance` method to reset the balance to zero and return the original balance, why does this method fail?

```
public int refundBalance() {  
    return balance;  
    balance = 0;  
}
```

6. How can we fix it?

Block Scope

7. In the method below, which lines will compile successfully?

```
public void scopeDemo(int param) {  
    int localVar = 5;  
    {  
        int innerVar = 10;  
        System.out.println(param + localVar + innerVar); // Line A  
    }  
    System.out.println(param + localVar); // Line B  
    System.out.println(innerVar); // Line C  
}
```

Lifetime

8. What is the lifetime of:

- a field?
- a local variable?

Thinking Question

9. Why does Java restrict access to variables outside their scope?

10. How does this help prevent bugs?